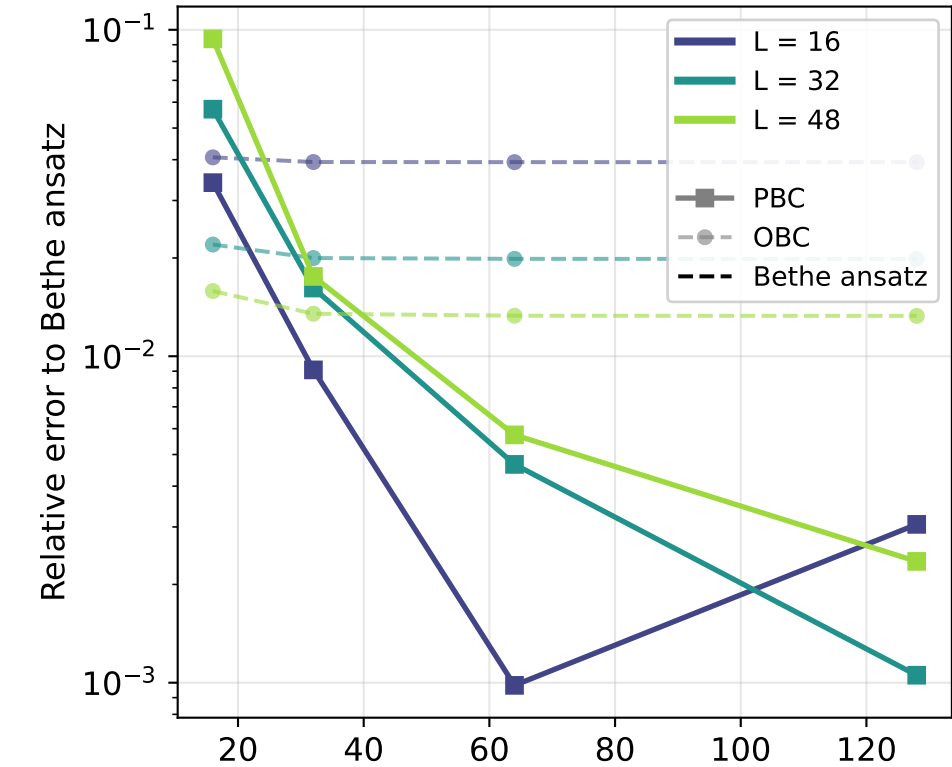
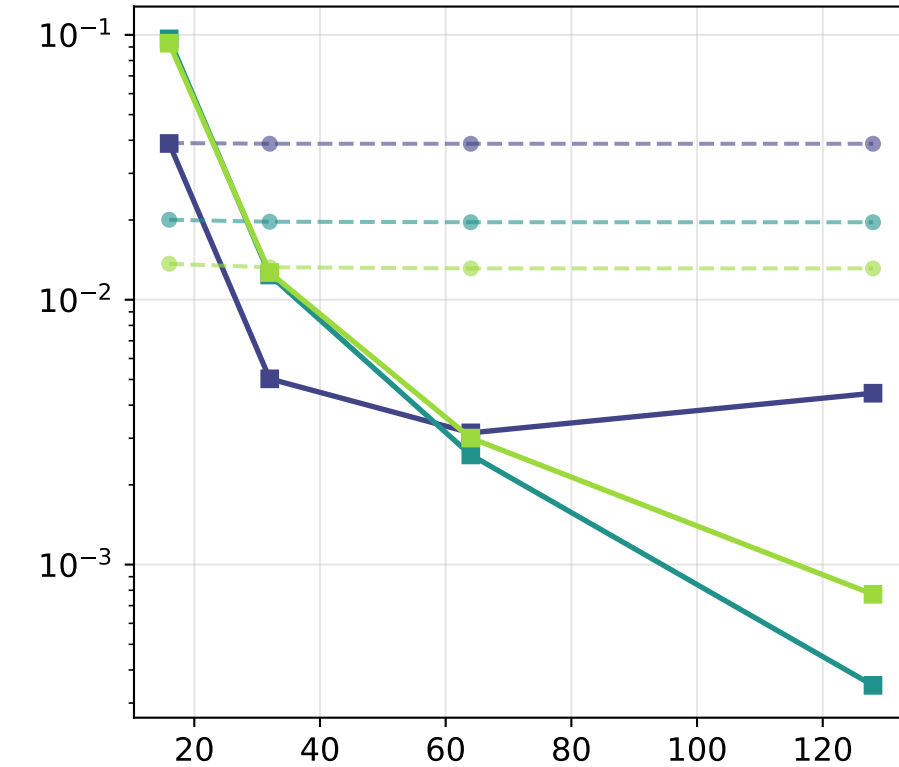
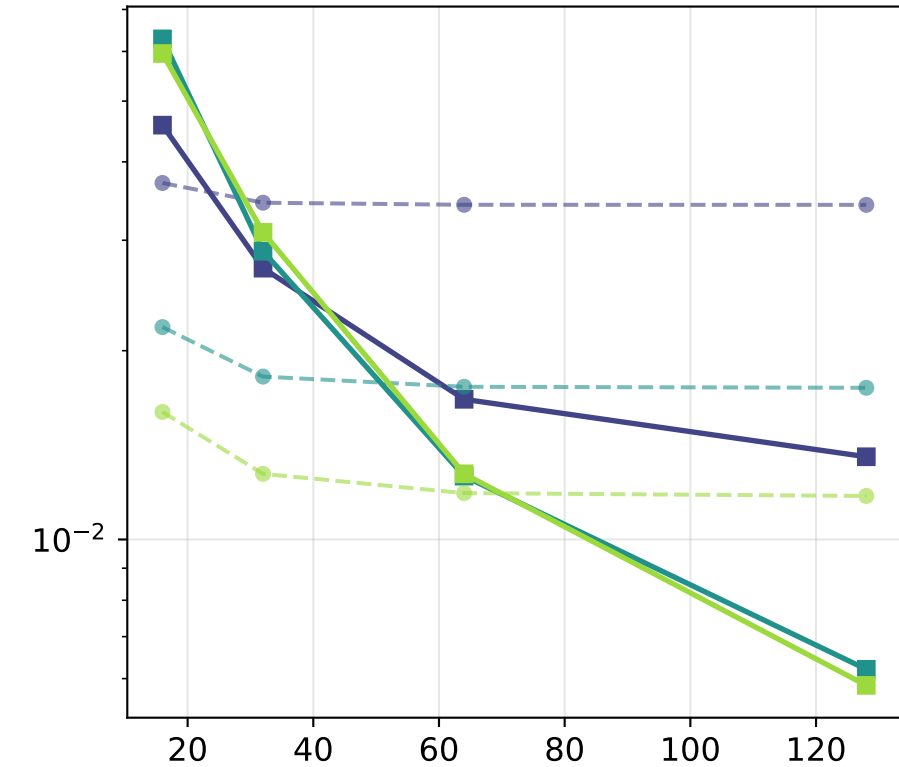


Half-fill $U=4$ (gapped)Half-fill $U=8$ (strongly gapped)Half-fill $U=0$ (free fermion)Quarter-fill $U=4$ (metallic)